

ECE

150-Day Complete Preparation Roadmap

Study Smart → Practice More → Be Consistent → Achieve Your Goal

"Discipline Today
Success Tomorrow" 😊

Target Exams

- ✓ GATE ECE
- ✓ ESE/IES - Electronics & Telecommunication
- ✓ PSU Technical Exams
- ✓ ISRO / DRDO / BEL / ECIL
- ✓ Other ECE technical recruitment exams

"Same Effort
Different Result
= Better Planning"

"Keep Going ...
You Can Do It!"



Why 150 Days?

- ✓ Complete syllabus systematically
- ✓ Concept building + problem practice
- ✓ Multiple revisions
- ✓ PYQs + Mock tests
- ✓ Boost Confidence for exam

Phase 1

Day 1 - 30

Foundation Building

1 Engineering Mathematics

- Linear Algebra
- Calculus
- Differential Equations
- Probability & Statistics
- Complex Variables
- Numerical Methods

2 Network Theory

- Basic Circuit Laws
- KCL & KVL
- Nodal & Mesh Analysis
- Network Theorems
- AC Circuits
- Resonance
- Transient Analysis
- Two-Port Networks
- Network Topology

3 Electronic Devices

- Semiconductor Physics
- PN Junction Diode
- Zener Diode
- BJT
- MOSFET
- JFET
- CMOS Basics
- Biasing
- Small-Signal Models

Weekly Target

6 days study + 1 day
revision/test

Phase 2

Day 31 - 70

Core ECE

4 Analog Circuits

- Diode Circuits
- BJT Amplifiers
- MOSFET Amplifiers
- Multistage Amplifiers
- Differential Amplifier
- Current Mirrors
- Feedback
- Oscillators
- Power Amplifiers

5 Digital Electronics

- Number Systems
- Boolean Algebra
- Logic Gates
- K-Maps
- Combinational Circuits
- Multiplexers/Demultiplexers
- Encoders/Decoders
- Flip-Flops
- Counters
- Registers
- Memory
- ADC/DAC

6 Signals & Systems

- Continuous-Time Signals
- Discrete-Time Signals
- LTI Systems
- Convolution
- Fourier Series
- Fourier Transform
- Laplace Transform
- Z-Transform
- Sampling Theorem

Concept → Formula →
20-30 Questions → PYQs
→ Error Analysis

Phase 3

Day 71 - 110

High-Weightage Subjects

7 Control Systems

- Transfer Function
- Block Diagram
- Signal Flow Graph
- Time Response
- Stability
- Routh-Hurwitz
- Root Locus
- Bode Plot
- Nyquist Plot
- State Space Analysis

8 Communication Systems

- Analog Communication
- AM, FM, PM
- Noise
- Sampling
- Pulse Modulation
- Digital Communication
- ASK / FSK / PSK
- QAM
- Information Theory

9 Electromagnetics

- Vector Algebra
- Coordinate Systems
- Coulomb's Law
- Electric Field
- Gauss Law
- Electric Potential
- Magnetic Field
- Maxwell's Equations
- Electromagnetic Waves
- Transmission Lines
- Waveguides
- Antennas

"Strong Concepts
Strong Results"

Phase 4

Day 111 - 130

Advanced + PYQ Mastery

10 Microprocessors & Microcontrollers

- 8085 Architecture
- 8086 Architecture
- Instruction Set
- Addressing Modes
- Memory Interfacing
- Interrupts
- 8051 Basics
- Timers/Counters
- Serial Communication

11 Digital Signal Processing

- Discrete-Time Signals
- DFT
- FFT
- Digital Filters
- FIR
- IIR
- Z-Transform Applications

Main Focus

- ✓ Last 20 years GATE PYQs
- ✓ ESE PYQs
- ✓ PSU-level questions
- ✓ Understand why wrong options are wrong
- ✓ Make short notes

"Practice PYQs
They Teach You
The Exam Mindset"

Phase 5

Day 131 - 150

Revision + Mock Tests

Day 131 - 135

- Revision:
 - Mathematics
 - Network Theory
 - Electronic Devices
 - Analog Circuits

Day 136 - 140

- Revision:
 - Digital
 - Signals & Systems
 - Control Systems

Day 141 - 145

- Revision:
 - Communication
 - Electromagnetics
 - DSP
 - Microprocessors

Day 146 - 150

- Full Mock Test Phase
 - Full GATE Mock
 - Full ESE Technical Mock
 - PYQ Mock
 - Formula Revision
 - Mistake Notebook
 - Weak Topic Revision

"Revise Solve Test
Repeat"



Weekly Study Pattern

- Monday : New Concepts
- Tuesday : New Concepts
- Wednesday : Concepts + Questions
- Thursday : New Concepts
- Friday : PYQs
- Saturday : Practice Test
- Sunday : Revision + Mistake Analysis



Daily Study Plan

6 Hours Plan

- 2 hr - Concept/Theory
- 2 hr - Numerical Problems
- 1 hr - PYQs
- 30 min - Formula Revision
- 30 min - Error Notebook

3 Hours Plan

- 1 hr - Concept
- 1 hr - Questions/PYQs
- 30 min - Revision
- 30 min - Formula + Mistakes



Priority Order for ECE

Very High Priority

1. Network Theory
2. Electronic Devices
3. Analog Circuits
4. Digital Electronics
5. Signals & Systems
6. Control Systems
7. Communication

High Priority

8. Engineering Mathematics
9. Electromagnetics
10. DSP

Additional

11. Microprocessors & Microcontrollers



Final Target (By Day 150)

- ✓ Complete ECE syllabus once
- ✓ 2-3 revisions
- ✓ 10-15 years GATE PYQs
- ✓ ESE PYQs
- ✓ 20+ subject tests
- ✓ 10+ full-length mocks
- ✓ Complete formula notebook
- ✓ Complete mistake notebook
- ✓ Regular numerical practice

"A Well Planned 150 Days
Can Change Your Future" 😊

Believe in Yourself
You Can Do It! 📈

PHASE 1 — Foundation Building

Build Strong Basics → Concepts → Practice → Confidence



Day 1 – 30

(First 30 Days)

Small Steps
Big Progress

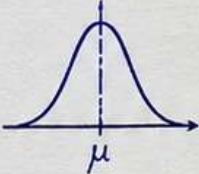


"A strong foundation today, a brighter ECE future tomorrow."

1 Engineering Mathematics

- Linear Algebra
- Calculus
- Differential Equations
- Probability & Statistics
- Complex Variables
- Numerical Methods

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \quad \int x^2 dx$$

$$\frac{dy}{dx} + y = 0$$


$$z = x + jy \quad f(x) \approx \frac{f(x+h) - f(x)}{h}$$

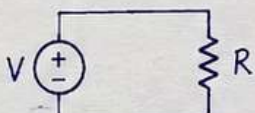
"Mathematics is the language of Engineering."

2 Network Theory

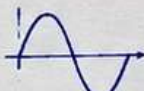
- Basic Circuit Laws
- KCL & KVL
- Nodal & Mesh Analysis
- Network Theorems
- AC Circuits
- Resonance
- Transient Analysis
- Two-Port Networks
- Network Topology

$$V = IR$$


$$\sum I_{in} = \sum I_{out} \quad (KCL)$$



$$\sum V = 0 \quad (KVL)$$

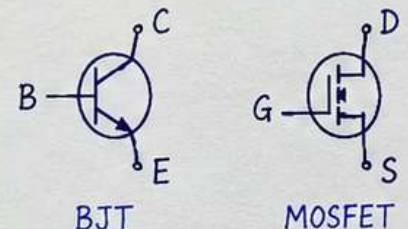
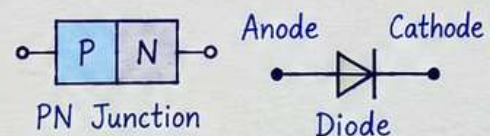


$$V(t) = V_m \sin(\omega t)$$

"Circuits are the building blocks of Technology."

3 Electronic Devices

- Semiconductor Physics
- PN Junction Diode
- Zener Diode
- BJT
- MOSFET
- JFET
- CMOS Basics
- Biasing
- Small-Signal Models



From Atoms to Applications

"Electronic Devices power the modern world."

Weekly Target:

6 days study + 1 day revision/test

- ✓ Understand concepts
- ✓ Solve problems
- ✓ Make short notes
- ✓ Attempt quizzes / PYQs
- ✓ Revise weekly

Mon	Tue	Wed	Thu	Fri	Sat	Sun
Study	Study	Study	Study	Study	Study	Revision / Test

"Discipline now, Success later."



Better Engineers Build a Better Tomorrow

PHASE 2 — Core ECE

(Build Core Knowledge → Strengthen Concepts → Solve → Excel)



Day 31-70

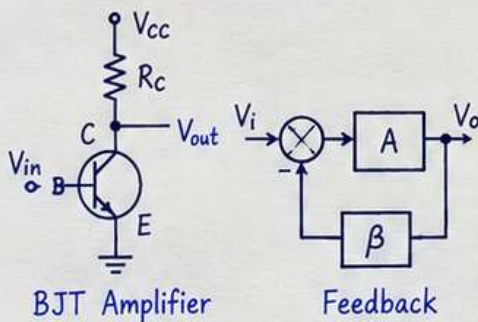
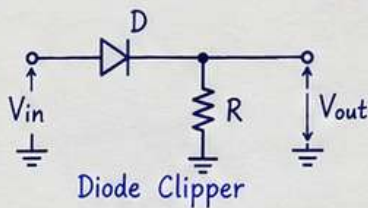
Deepen Understanding
Build Core Skills



"Concepts today, Expertise tomorrow."

4 Analog Circuits

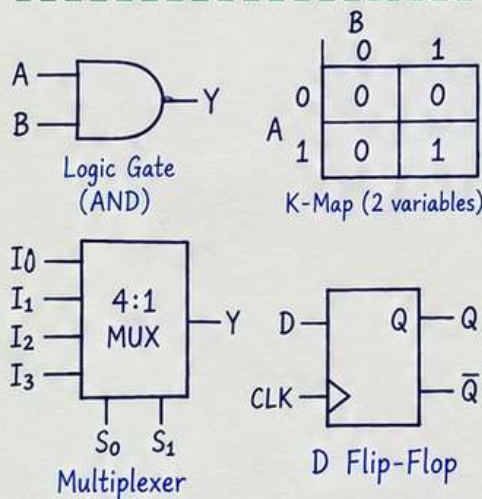
- Diode Circuits
- BJT Amplifiers
- MOSFET Amplifiers
- Multistage Amplifiers
- Differential Amplifier
- Current Mirrors
- Feedback
- Oscillators
- Power Amplifiers



"Analog Circuits turn signals into possibilities."

5 Digital Electronics

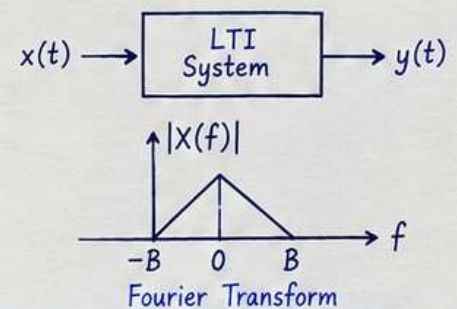
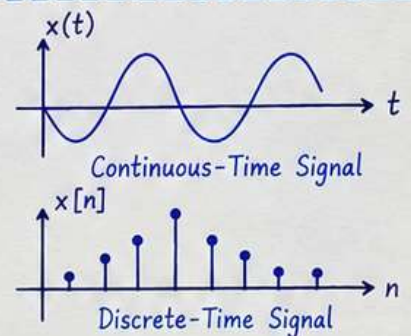
- Number Systems
- Boolean Algebra
- Logic Gates
- K-Maps
- Combinational Circuits
- Multiplexers/Demultiplexers
- Encoders/Decoders
- Flip-Flops
- Counters
- Registers
- Memory
- ADC/DAC



"Digital Electronics builds the brain of modern systems."

6 Signals & Systems

- Continuous-Time Signals
- Discrete-Time Signals
- LTI Systems
- Convolution
- Fourier Series
- Fourier Transform
- Laplace Transform
- Z-Transform
- Sampling Theorem



"Signals & Systems connect the time domain to the frequency domain."

Practice Rule (For Every Topic)

Concept
(Understand the basics)



Formula
(Learn key formulas)



20-30 Questions
(Practice well)



PYQs
(Solve previous year questions)



Error Analysis
(Find & fix mistakes)

"Practice with purpose, analyse your mistakes, and watch your progress compound."



Weekly Target: 6 days study + 1 day revision/test

"Discipline in Phase 2 creates Confidence for Phase 3."

"Focus on what truly matters."

PHASE 3 — High-Weightage Subjects

Deeper Concepts → Practice → Problem Solving → Mastery



Day 71 – 110

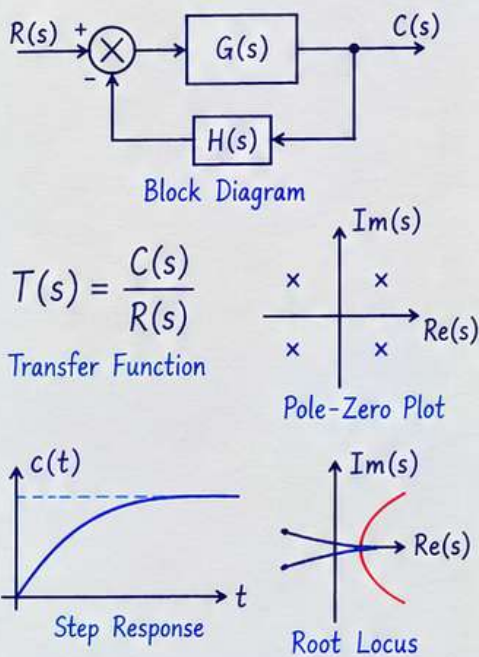
"High-Weightage Today, High Opportunities Tomorrow."



"Hard Topics Build Strong Engineers."

7 Control Systems

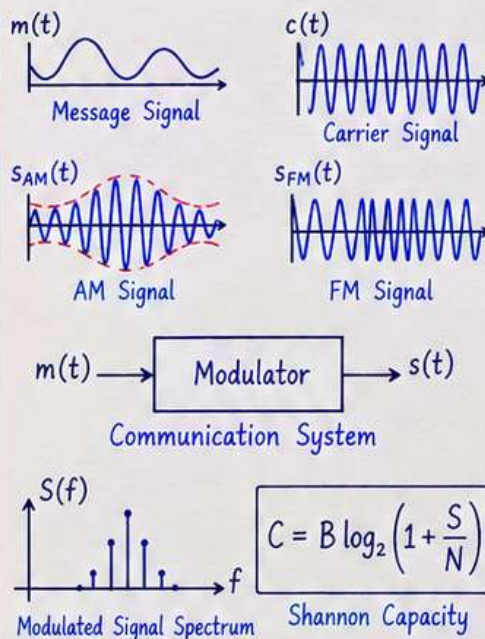
- Transfer Function
- Block Diagram
- Signal Flow Graph
- Time Response
- Stability
- Routh-Hurwitz
- Root Locus
- Bode Plot
- Nyquist Plot
- State Space Analysis



"Control Systems bring stability to real-world systems."

8 Communication Systems

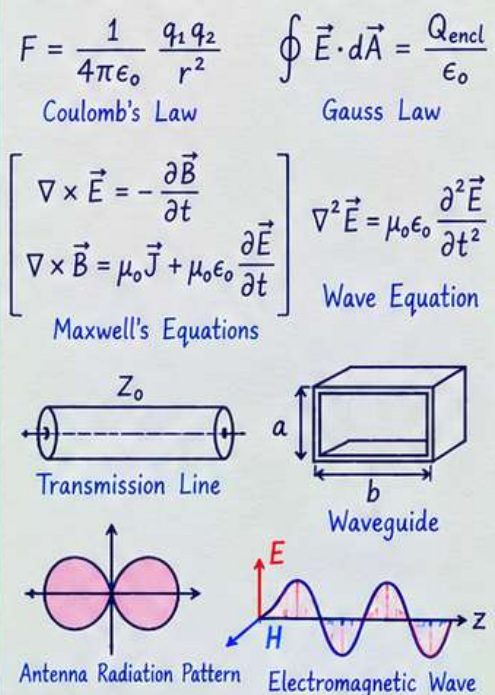
- Analog Communication
- AM
- FM
- PM
- Noise
- Sampling
- Pulse Modulation
- Digital Communication
- ASK / FSK / PSK
- QAM
- Information Theory



"Communication Systems connect the world."

9 Electromagnetics

- Vector Algebra
- Coordinate Systems
- Coulomb's Law
- Electric Field
- Gauss Law
- Electric Potential
- Magnetic Field
- Maxwell's Equations
- Electromagnetic Waves
- Transmission Lines
- Waveguides
- Antennas



"Electromagnetics is the foundation of modern wireless technology."

Practice Rule (For Every Topic)

Concept
(Understand the basics)



Formula
(Learn key formulas)



20-30 Questions
(Practice well)



PYQs
(Solve previous year questions)



Error Analysis
(Find & fix mistakes)

"Practice smart, not just hard."



Weekly Target: 6 days study + 1 day revision/test

"Discipline in Phase 3 creates opportunities in Phase 4."

"Past Questions Build Future Selections."

PHASE 4 – Advanced + PYQ Mastery

Practice Deeper → Solve PYQs → Analyze → Improve → Excel



Day 111 – 130

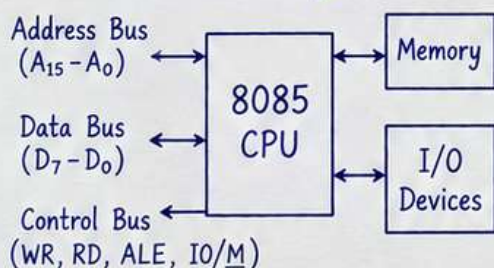
"It's not just about solving, it's about understanding."

"Analyze Learn Improve Repeat."

10 Microprocessors & Microcontrollers

- 8085 Architecture
- 8086 Architecture
- Instruction Set
- Addressing Modes
- Memory Interfacing
- Interrupts
- 8051 Basics
- Timers/Counters
- Serial Communication

8085 (Block Diagram)



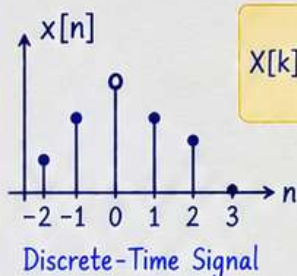
8051 (Pin Diagram)



"Microprocessors give the brain; Microcontrollers bring it to life."

11 Digital Signal Processing

- Discrete-Time Signals
- DFT
- FFT
- Digital Filters
- FIR
- IIR
- Z-Transform Applications

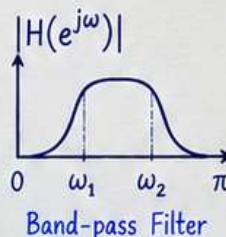
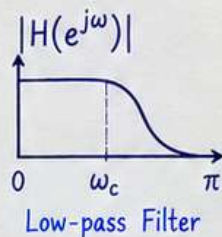


$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi kn}{N}}$$

DFT

$$X[k] = \text{FFT}\{x[n]\}$$

Fast Fourier Transform



"DSP turns signals into insights."

Main Focus

- ✓ Last 20 years GATE PYQs
- ✓ Last 20 years ESE PYQs
- ✓ PSU-level questions
- ✓ Topic-wise + Subject-wise practice
- ✓ Mix of conceptual, numerical and theory questions

Don't just solve PYQs. Understand why the wrong options are wrong."

PYQ Solving Strategy

- 1 Solve the question
- 2 Analyze all options (correct & incorrect)
- 3 Identify the concept used
- 4 Make short notes of important questions
- 5 Track mistakes
- 6 Revise regularly

"Every PYQ is a lesson, not just a question."



Daily Plan (Day 111 – 130)

Concept Revision	PYQ Practice	Analysis & Short Notes	Mock Test (Alternate Days)
2-3 hrs	2-3 hrs	1-2 hrs	1-2 hrs



Your Target

- ✓ Conceptual Clarity
- ✓ Higher Accuracy
- ✓ Better Time Management
- ✓ Build Exam Temperament

"Solve Learn Analyze Improve Succeed."

"Same Questions, Different Mindset, Better Results!"

"Revise
Sharpen
Test
Succeed."

PHASE 5 — Revision + Mock Tests

Review What You Learned → Practice Like an Exam → Build Final Confidence



Day 131 – 150

"Revision converts
knowledge into results."

"Mock
today
Stronger
tomorrow."

Days 131 – 135

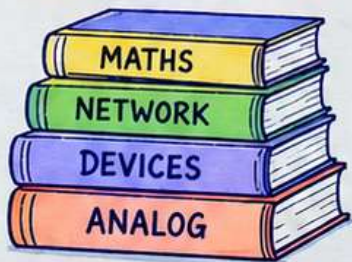
Revision :

- Mathematics
- Network Theory
- Electronic Devices
- Analog

Focus Points:

- ✓ Revise formulas
- ✓ Go through short notes
- ✓ Solve important questions
- ✓ Practice previous year questions

"Strong Basics
make a Strong
Finish."



Days 136 – 140

Revision :

- Digital
- Signals & Systems
- Control

Focus Points:

- ✓ Revise key concepts and formulas
- ✓ Practice numerical and theory questions
- ✓ Solve PYQs
- ✓ Make short notes / cheat sheet

"Practice today
for a better
tomorrow."



Days 141 – 145

Revision :

- Communication
- Electromagnetics
- DSP
- Microprocessors

Focus Points:

- ✓ Revise important derivations
- ✓ Solve conceptual + application based Qs
- ✓ Practice PYQs (Last 10-20 years)
- ✓ Improve speed and accuracy

"Revise Deeply
Solve Smartly
Score Highly."



Days 146 – 150

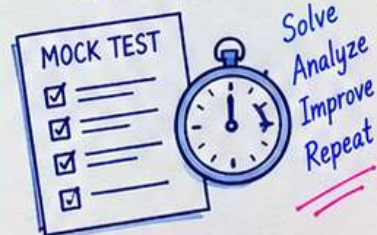


Full Mock Test Phase

- Full GATE Mock
- Full ESE Technical Mock
- PYQ Mock
- Formula Revision
- Mistake Notebook
- Weak Topic Revision

How to Do It?

- ✓ Give mocks in exam environment (timer)
- ✓ Analyze performance (accuracy + time)
- ✓ Maintain mistake notebook
- ✓ Revise weak topics
- ✓ Focus on improving each mock score



"Tests don't show
your weakness, they show
your next target."



Daily Routine (Phase 5)

Revision (Concepts + Notes)	Practice Questions (PYQs / Subject-wise)	Mock Test (Alternate Days)	Analysis & Weak Topic Revision
3-4 hrs	2-3 hrs	2-3 hrs	1-2 hrs



Final Target

- ✓ Be exam ready
- ✓ Improve accuracy
- ✓ Strengthen weak areas
- ✓ Build confidence
- ✓ Maximize your score

"Discipline
in Revision
delivers
Success
in Exam."

"Same Effort + Smarter Revision = Bigger Results."

"A Clear Plan Brings Big Results."

STUDY PLAN & STRATEGY

Plan Well → Study Smart → Solve More → Crack It!

"Discipline today, Dream job tomorrow."



Weekly Study Pattern

Day	Work
Monday	New Concepts
Tuesday	New Concepts
Wednesday	Concepts + Questions
Thursday	New Concepts
Friday	PYQs
Saturday	Practice Test
Sunday	Revision + Mistake Analysis



"Be consistent. One week at a time."



Daily Study Plan

If you can study 6 hours/day:

- 2 hr → Concept / Theory (Read + Make Notes)
- 2 hr → Numerical Problems (Practice & Apply)
- 1 hr → PYQs (Solve Previous Year Questions)
- 30 min → Formula Revision (Key formulas + Short notes)
- 30 min → Error / Mistake Notebook (Analyze & Learn)

If you have only 3 hours/day:

- 1 hr → Concept (Read + Understand)
- 1 hr → Questions / PYQs (Practice)
- 30 min → Revision (Quick Summary)
- 30 min → Formula + Mistakes (Revise + Update Notebook)



The Golden Strategy

First 50 Days Learn

- Build fundamentals
- Cover all core topics
- Make short notes
- Solve basic questions



Next 60 Days Learn + Solve

- Strengthen concepts
- Solve more numerical questions
- Practice subject-wise PYQs
- Improve speed and accuracy



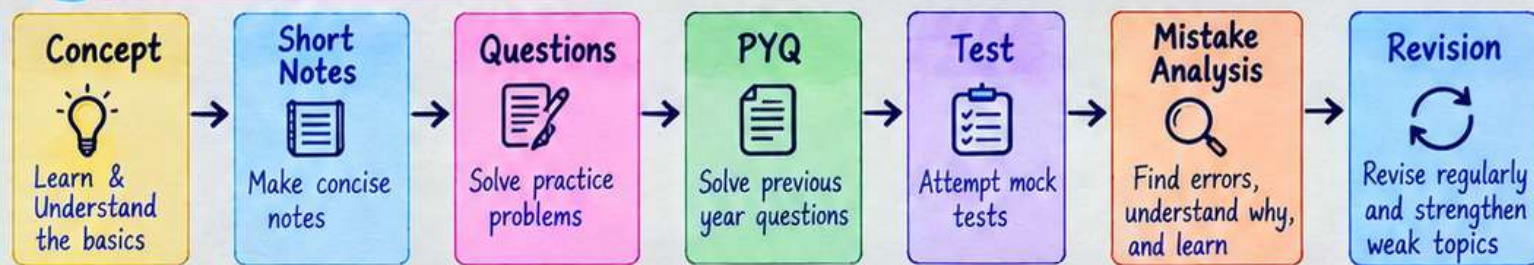
Last 40 Days Revise + PYQs + Mock Tests

- Complete revision
- Solve last 20 years PYQs
- Attempt full-length mock tests
- Focus on weak topics
- Analyze mistakes and improve

"Same Effort + Smarter Strategy = Bigger Results."



Your Preparation Cycle



Pro Tips

- 1 Be consistent (No long breaks)
- 2 Make short notes (Revise easily)
- 3 Solve more and more questions
- 4 Analyze mistakes honestly
- 5 Take mock tests in exam environment
- 6 Stay positive and keep improving

"It's not just about studying hard, It's about studying smart."



Remember:

- ✓ Small daily progress adds up
- ✓ Quality > Quantity
- ✓ Practice is the real preparation
- ✓ Mistakes are opportunities to learn
- ✓ Stay disciplined and keep going

"A better you builds a brighter tomorrow."

"Don't Study Hard, Study Smart."

Priority Order for ECE

"Right Priorities Lead to Big Success."

Focus Smart → Study Deep → Practice More → Get Results



Very High Priority

(Must Master)

- 1 Network Theory (Core + Many Questions)
- 2 Electronic Devices (Fundamental)
- 3 Analog Circuits (High Weightage)
- 4 Digital Electronics (Conceptual + Scoring)
- 5 Signals & Systems (Important for GATE/ESE)
- 6 Control Systems (Numerical Intensive)
- 7 Communication (Concept + Numerical)



High Priority

(Very Important)

- 8 Engineering Mathematics (Base for Everything)
- 9 Electromagnetics (Conceptual + Moderate Numericals)
- 10 DSP (Growing Importance)



Additional

(Do Not Ignore)

- 11 Microprocessors & Microcontrollers (Scoring + Easy to Complete)

"Prioritize, Plan and Perform – That's the Winning Formula!"



150-Day Final Target

"Discipline Today Dream Career Tomorrow."



- ✓ Complete ECE syllabus once
- ✓ 2 – 3 complete revisions
- ✓ 10 – 15 years GATE PYQs
- ✓ ESE PYQs
- ✓ 20+ subject/topic tests

- ✓ 10+ full-length mocks
- ✓ Complete formula notebook
- ✓ Complete mistake notebook
- ✓ Regular numerical practice
- Be confident and exam ready!



Small Consistent Efforts Lead to Big Results.

"150 Days of Focus can change your future."



Same Syllabus Different Mindset Bigger Results!

CONSISTENCY + SMART WORK + PRACTICE = SUCCESS

You Can Do It!